

Hybrid Quantum-Classical Framework for Computational Mental Energy from Multichannel EEG Streams

A standardized cognitive indicator (CME, Vernik unit) with hybrid quantum-classical inference, validated on real IBM Marrakesh hardware

Mykhailo Vernik, Liubov Oleshchenko, Zhengbing Hu

Igor Sikorsky Kyiv Polytechnic Institute · Hubei University of Technology

ICSSEA 2026 · Article No. 2614

Why this matters

The opportunity

- Wearable EEG is now portable, real-time, low-friction
- Adaptive HCI, learning, neurofeedback, workplace cognition all need *operational* cognitive metrics
- Quantum ML is moving from offline benchmarks toward streaming use

Three persistent gaps

1. Pipelines fragment feature extraction and decisions; no unified output unit
2. Quantum ML for EEG is evaluated offline, not in production-grade streams
3. Quantum-resource cost is rarely modelled together with prediction quality

Contributions

1. **Computational Mental Energy (CME)** — a window- and session-level cognitive indicator with an explicit unit, the **Vernik** ($1 \text{ Vn} \equiv 1 \mu V^2 \cdot s$).
2. **Hybrid quantum-classical inference** — a 4-qubit data-re-uploading VQC fused with a classical MLP via a single mixing weight μ .
3. **Resource-aware metaheuristic optimization** — jointly tunes model parameters Θ , shots S , depth D against quality + cost.
4. **Deployable streaming architecture** — wearable EEG $\rightarrow$ edge $\rightarrow$ server, with quantum-only / classical-only / hybrid backends behind one CME formalism.

EEG Quantum ML Streaming Real QPU validation

Computational Mental Energy (CME)

Idea. Combine *what the EEG shows* (spectral energy), *what the user is doing* (task complexity c), and *how engaged they are* (flow probability p_{flow}) into one comparable number per window.

Aggregated band power per window:

$$E_{\text{band}}(t) = \sum_{ch \in \mathcal{CH}} (w_{\delta} P_{\delta} + w_{\theta} P_{\theta} + w_{\alpha} P_{\alpha} + w_{\beta} P_{\beta} + w_{\gamma} P_{\gamma})$$

Modulation by complexity and flow:

$$g(c, p) = \lambda_1 c + \lambda_2 p + \lambda_3 cp$$

CME rate and window CME:

$$\text{CME}_{\text{rate}}(t) = \kappa \cdot E_{\text{band}}(t) \cdot g(c(t), p_{\text{flow}}(t)), \quad \text{CME}(t) = \text{CME}_{\text{rate}}(t) \cdot \Delta$$

Three knobs (energy, complexity, engagement) → one number per window.

CME — worked example (one 5-s window of *reading technical*)

Inputs

- Aggregated band power: $E_{\text{band}} = 4.9903 \mu V^2$
- Task complexity: $c = 0.62$ (reading technical)
- VQC over 1024 shots: $p_{\text{flow}} = 0.623$
- Default weights: $\lambda_1 = \lambda_2 = \lambda_3 = 0.5, \kappa = 1$

Modulation

$$g(c, p) = 0.5 \cdot 0.62 + 0.5 \cdot 0.623 + 0.5 \cdot 0.62 \cdot 0.623$$

$$= 0.310 + 0.312 + 0.193 = \mathbf{0.815}$$

Compose

$$\text{CME}_{\text{rate}} = 1.0 \cdot 4.99 \cdot 0.815 = \mathbf{4.07 \text{ Vn/s}}$$

$$\text{CME}(\Delta = 5 \text{ s}) = 4.07 \cdot 5 = \mathbf{20.33 \text{ Vn}}$$

Why a new unit (Vernik)?

Definition

- $1 \text{ Vn} \equiv 1 \mu\text{V}^2 \cdot \text{s}$
- Time-integral of squared EEG amplitude over a window Δ
- Pure signal-energy unit (no impedance assumption)

Why a named symbol?

- Cognitive expenditure is reported across sessions, devices, dashboards
- A single named magnitude is operationally cleaner — same logic as **kWh** naming a derived joule magnitude
- *Not a new physics claim; an analytics convention*

Concrete intuition — at the rates measured in our pilot:

Streaming architecture (6 layers)

Pipeline

1. Acquisition — wearable EEG (Muse Athena, 4 ch)
2. Ingestion — windowing, quality filter, normalization
3. Features — Welch PSD per band; full $\mathbb{R}^{22}$ + reduced $\mathbb{R}^8$
4. Inference — quantum / classical / hybrid backend
5. CME engine — $\text{CME}_{\text{rate}}(t)$, $\text{CME}(t)$, session aggregates
6. Persistence + Web API (async)

Closed-loop optimizer

- Background metaheuristic (GA / PSO / ACO / SA)
- Tunes (Θ, S, D) against quality vs cost

A window's journey — ~1.59 s end-to-end

Step	Median latency	Where
EEG window arrival ($\Delta = 5\text{ s}$)	$t = 0$	edge node
Ingest, quality check, band-pass filter	~30 ms	server
Welch PSD $\times$ 4 channels $\times$ 5 bands	~80 ms	server
Feature transforms ($\mathbb{R}^{22}, \mathbb{R}^8$)	~10 ms	server
Quantum inference (1024 shots)	~1456 ms	Aer / QPU
Classical NN forward pass	~5 ms	server
Fusion + CME engine	< 1 ms	server
Async persistence + API response	~10 ms	server
End-to-end (reported)	~1589 ms	—

Quantum dominates latency. The metaheuristic optimizer trades shots/depth for speed when budgets are tight; classical fallback engages if QPU is unavailable.

4-qubit Variational Quantum Classifier

Per layer $\ell \in \{0, \dots, L\}$:

1. **Dual-axis re-uploading** — $R_y((x_i + 1)\pi), R_z((x_{i+4} + 1)\pi)$
2. **Feature interaction** — ring-wise R_{ZZ} between adjacent qubits
3. **Entanglement** — CNOT ring $q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3 \rightarrow q_0$
4. **Trainable** — $R_y(\theta_i^{(\ell)}), R_z(\phi_i^{(\ell)})$

Compactness: $N_q = 4, L = 2 \Rightarrow |\Theta| = 2N_q(L + 1) = \mathbf{24}$ trainable parameters.

Readout: $p_{\text{flow}}(t) = \Pr[b_0 = 1]$ on q_0 .

Re-uploading at every layer increases the expressive power of an otherwise shallow circuit — important when only 4 qubits are available.

VQC encoding — a worked example

Reduced feature vector $x \in \mathbb{R}^8$ (band averages + asymmetry + engagement + complexity), normalized to $[-1, 1]$:

$$x = [0.30, 0.50, 0.20, 0.40, 0.10, 0.60, 0.70, 0.50]$$

Qubit	R_y angle	R_z angle	Encodes
q_0	$(0.30 + 1)\pi = 1.30\pi$	$(0.10 + 1)\pi = 1.10\pi$	$\bar{P}_\delta$ + frontal asymmetry α
q_1	$(0.50 + 1)\pi = 1.50\pi$	$(0.60 + 1)\pi = 1.60\pi$	$\bar{P}_\theta$ + engagement β/θ
q_2	$(0.20 + 1)\pi = 1.20\pi$	$(0.70 + 1)\pi = 1.70\pi$	$\bar{P}_\alpha$ + task complexity $c(t)$
q_3	$(0.40 + 1)\pi = 1.40\pi$	$(0.50 + 1)\pi = 1.50\pi$	$\bar{P}_\beta$ + (reserved)

Per layer: R_{ZZ} ring $\rightarrow$ CNOT ring $\rightarrow$ trainable $R_y(\theta), R_z(\phi)$.

Measurement: 1024 shots on $q_0 \rightarrow \text{count}(b_0 = 1) = 638 \rightarrow p_{\text{flow}} = 0.623$.

Hybrid fusion

$$p_{\text{flow}}^{\text{hybrid}}(t) = \mu \cdot p_{\text{flow}}^{\text{Q}}(t) + (1 - \mu) \cdot p_{\text{flow}}^{\text{NN}}(t), \quad \mu \in [0, 1]$$

Three modes, one CME core

- $\mu = 1$ — quantum-only
- $\mu = 0$ — classical-only
- $\mu = 0.6$ — recommended production

Branches in parallel

- Quantum: 4-qubit VQC on $\mathbb{R}^8$ reduced features
- Classical: MLP ($22 \rightarrow 64 \rightarrow 32 \rightarrow 1$) on full $\mathbb{R}^{22}$

Fusion happens *before* the CME engine, so the energy formula never changes. μ is a *trust knob*: it can be tuned from data or set by policy.

Hybrid fusion — a worked example

Same window, two estimators

- Quantum: $p_{\text{flow}}^Q = 0.85$ — high confidence, but per-window variance $\sigma_Q^2 = 0.0140$
- Classical: $p_{\text{flow}}^{NN} = 0.42$ — less peaked, lower variance $\sigma_{NN}^2 \approx 0.005$

Apply $\mu = 0.6$:

$$p_{\text{flow}}^{\text{hybrid}} = 0.6 \cdot 0.85 + 0.4 \cdot 0.42 = 0.510 + 0.168 = \mathbf{0.678}$$

Variance after fusion (treating branches as approximately independent):

$$\sigma_{\text{hybrid}}^2 \approx \mu^2 \sigma_Q^2 + (1 - \mu)^2 \sigma_{NN}^2$$

In our pilot this drops the empirical p_{flow} variance from **0.0140** → **0.0083** — a **−40.9 %** reduction, so flow predictions stop oscillating between back-to-back windows and the cumulative CME trace becomes smooth enough for live dashboards.

Experimental setup

Single-subject pilot

- Muse Athena headband (TP9, AF7, AF8, TP10)
- 8 cognitive activities (rest → coding/math)
- 288 five-second windows, 24 min recording
- Train/eval split with held-out evaluation subset
- Weak labels heuristically refined (FSS / FKS to follow)

Backends

- Aer simulator (1024 shots/window)
- **IBM Marrakesh** — 156-qubit Heron r2 (real QPU)
- 100-window stratified sample submitted as one batch job

Reported numbers are point estimates; no inter-subject confidence intervals are claimed.

Results — flow-state classification

Mode	Accuracy	F1	AUROC
Quantum-only VQC (4 qubits, 24 params)	0.500	0.153	0.548
Hybrid ($\mu = 0.6$)	0.882	0.553	0.914

- Hybrid reduces p_{flow} variance **–40.9 %** vs quantum-only (0.0140 → 0.0083).
- F1 / accuracy gap reflects class imbalance in the small evaluation set.
- 24 trainable parameters is *intentional* — small enough to train on a single subject without overfitting and to fit on a 4-qubit chip.

Single-subject pilot. Cross-subject validation is the first item of future work; recent wearable-EEG flow studies provide LOSO benchmarks to compare against [Beiramvand et al., *Proc. EMBC 2025*: 93 % within-subject, 82 % LOSO over 29 subjects].

Results — CME per activity

Activity	$c(t)$	p_{flow}	CME rate (Vn/s)
Coding	0.70	0.456	339.9
Math / Problem solving	0.90	0.442	316.7
Debugging	0.80	0.469	277.0
Reading (technical)	0.60	0.484	190.5
Reading (general)	0.35	0.472	149.3
Email	0.30	0.415	132.5
Browsing	0.20	0.484	112.5
Resting	0.05	0.551	37.1

Take-aways

- ~9× rate ratio between coding and rest
- Super-linear with $c(t)$ in the fitted curve

• 9.5-h day extrapolates to $\approx 7.6 \text{ M Vn}$

• *Single-subject illustration* — not a population

Results — IBM Marrakesh validation

Hardware run

- 100 stratified EEG windows (≈ 12 -13 per activity)
- 1 batch job, 55.5 s on real QPU
- Transpiled depth: 61 $\rightarrow$ 219 ($\times 3.6$ routing overhead)
- Heron r2 routing is already an improvement over Eagle ($\times 4.4$)

Simulator vs real hardware

- Pearson $r = 0.869$, MAE = 0.045
- p_{flow} std: sim 0.103 $\rightarrow$ QPU 0.068
- Hybrid accuracy: sim 91 % $\rightarrow$ QPU 96 %
- Per-activity sim-vs-QPU disagreement < 0.034 (worst: Email, Resting)

Limitations (stated up front)

- **Sample size.** $N = 1$ subject, 288 windows; reported AUROC / accuracy / r are point estimates without inter-subject confidence intervals.
- **Labels.** Weak labels from a classical NN, refined heuristically; not yet validated against FSS / FKS on this dataset.
- **Hardware sample.** Single 100-window batch on one QPU; results are indicative across providers and sessions.
- **SI mapping.** $\text{CME} \rightarrow J$ assumes a simple electrode impedance — an approximation, not a physical claim.
- **Sleep / restoration model (§ 6.4).** Conceptual extension only; *not validated experimentally* in this paper.

Future work

- Multi-subject empirical validation, FSS / FKS calibration, transfer learning baselines
- Longitudinal CME balance & restoration model (multi-day, sleep tracking)
- Error mitigation on Heron — Dynamical Decoupling, ZNE, PEC
- Online policy learning for S , D , and μ under latency budgets
- Cross-device generalization (Muse 2, Muse S, Emotiv, Neurosity)
- Recent work to build on: hybrid quantum-classical EEG frameworks for clinical signals [Padmaja et al., *Sci. Rep.* 2026]

Take-aways

1. **CME + Vernik** give EEG-derived cognitive expenditure a single named magnitude — comparable across sessions, devices, subjects.
2. **Hybrid fusion** at $\mu = 0.6$ cuts flow-probability variance by **40.9 %** and reaches **0.914 AUROC** on a single-subject pilot.
3. **Real QPU run** on IBM Marrakesh agrees with the simulator at $r = 0.869$, $MAE = 0.045$ — feasibility on Heron-class hardware *today*.
4. **Open questions** are explicit: multi-subject validation, longitudinal restoration, error mitigation, and policy learning for resource control.

A reproducible streaming pipeline + a measurable cognitive unit + a working real-QPU path — all behind one CME formalism.

Thank you — questions?

Mykhailo Vernik

mykhailo.vernik@pzks.fam.kpi.ua

ORCID: 0009-0008-6156-1051

Liubov Oleshchenko · Zhengbing Hu

ICSSEA 2026 — Article No. 2614